

Bootstrap/Jackknife

Math 742

2026-01-07

Resampling Methods

Resampling methods involve resampling the training data and then refitting the model in order to gain more information about the fitted model. The basic procedure for resampling is:

- Drawing repeated samples from the training data.
- For each new sample, refit the model.
- Examine all of the models in order to develop your conclusions.

Three basic methods involving resampling are:

1. Cross-validation
2. Bootstrap
3. Jackknife

Key Idea: In classical statistics, we derive the sampling distribution analytically. In resampling methods, we approximate the sampling distribution computationally using the data itself.

Cross validation is typically used as a way to estimate the test error associated with some statistical learning method as a way to estimate its performance or to select the level of flexibility.

Bootstrapping is used to approximate the sampling distribution of an estimator, which then allows estimation of standard errors, bias, and confidence intervals for a parameter.

Jackknife is used to estimate the standard error and bias of a statistic, especially when the theoretical distribution of the estimator is difficult to derive, or the sample size is moderate and resampling is feasible.

Recall that we can reduce the MSE of our training set by using a more flexible model. However, if we use a more flexible the model this will result in a higher testing set MSE. Cross-validation is used to get a good handle on the test set error, whereas bootstrap is used to get a decent estimate of the variability or standard deviation.

This lecture will discuss that bootstrap method and the jackknife. Later on we will cover cross-validation.

Bootstrap

Bootstrap Basics and History

In statistics, we typically make assumptions about the population at large. We might for example assume the population follows a normal distribution. However, if we have no preconceived thoughts regarding the population or if previous experience dictates that the population does not follow some specified distribution we may have to resort to non-parametric statistics. The empirical bootstrap assumes the sample is iid, and the empirical distribution function $\hat{F}_n$ is a good approximation to F . The empirical bootstrap is nonparametric, since it replaces the unknown population distribution F with the empirical distribution $\hat{F}_n$, rather than assuming a parametric family.

Bootstrapping is a procedure developed by Bradley Efron at Stanford <https://statistics.stanford.edu/about/history/software-tools#:~:text=Bootstrap%20technique%20was%20invented%20by,up%20by%20your%20own%20bootstraps>).

The empirical “bootstrap” is the distribution of the statistic based on resampling the original data. Suppose I rolled a 6-sided die 5 times. The “resample” resamples the data “with replacement. This results in a second sample which because replacement is done, the same value can occur more than once.

Here is a quick example of resampling. First we “pretend” to roll a 6-sided die five times. This gives us

```
set.seed(8000)
dicerolls <- sample(1:6,5,replace=TRUE)
dicerolls

## [1] 6 3 4 3 1

resample <- sample(dicerolls,5,replace=TRUE)
resample

## [1] 4 4 3 6 4
```

Notice that because we are sampling with replacement in the resample, the same values can occur more than once (e.g. 4), and some values are never selected (e.g. 1). Finally, in resampling, you will never get something that wasn’t in your original dataset.

The bootstrap principle

- Original sample: $x_1, x_2, x_3, \dots, x_n$ and F is the distribution from which the sample is drawn.
- Original statistic from the sample: u , where u can be the sample mean, proportion etc.
- Resample (of same size): $x'_1, x'_2, x'_3, \dots, x'_n$ and F' is the distribution of the resample.
- Bootstrap statistic: u' is computed from the resample.
- The distribution F' approximates the true distribution F .
- The statistic u' approximates the sampling distribution of u .
- The variation of u can be approximated by the variation of u' .

The resample size is taken to be the same as the original sample size so that the bootstrap distribution mimics the sampling distribution of the estimator computed from samples of size n .

Bootstrapping is amazing because it gives us a sense of the variation of our point estimate. It is justified by the plug-in principle: **replace unknown population quantities with their empirical counterparts and study the induced distribution of the estimator.**

The LLN explains why the empirical distribution converges to the population distribution, but proving bootstrap consistency requires stronger results such as the **Glivenko–Cantelli theorem** and the functional delta method.

Bootstrap Simulation Experiment

For this experiment, we assume we have a normally distributed population whose mean is 10 and standard deviation is 3.

I randomly select 30 items from the population.

```
set.seed(1500)
xsample <- rnorm(30,mean=10,sd=3)
```

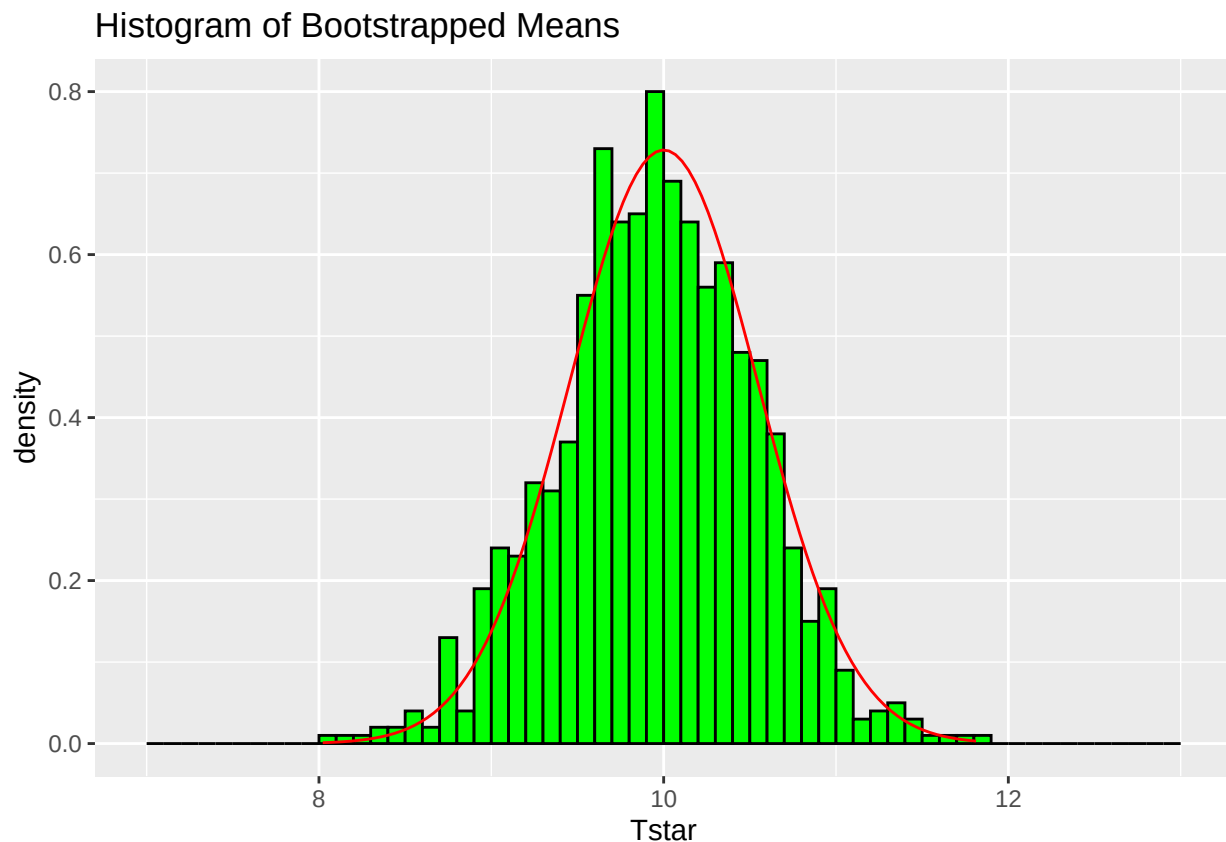
Suppose we wished to determine the bootstrap estimate of the mean. In order to do this, we take 1000 samples from our original sample (i.e. xsample) with replacement → this is the bootstrap principle.

For each sample we calculate the mean.

```
Tstar <- matrix(NA,nrow=1000) # initialize matrix to hold means
for (i in 1:1000){
  bsample <- sample(xsample, size = length(xsample), replace = TRUE)
  Tstar[i] <- mean(bsample) # calculate mean of bootstrap sample
}
```

Visualize the results by creating a histogram of the means.

```
tibble(Tstar) %>% ggplot(aes(x = Tstar)) +
  geom_histogram(aes(y =after_stat(density)),
    breaks = seq(7, 13, by = 0.1),
    color = "black",
    fill = "green") +
  stat_function(fun = dnorm, col="red",args = list(mean = 10, sd = 3/sqrt(30))) +
  labs(title = "Histogram of Bootstrapped Means")
```



Since this is a made up example, we know that the mean of the population is 10. If we had just one sample and then “bootstrapped” that sample 1000 times we can now approximate the sampling distribution of the sample mean and therefore estimate its standard error and confidence interval.

We can also create summary statistics for the bootstrapped means.

```
favstats(Tstar)
```

##	min	Q1	median	Q3	max	mean	sd	n missing
----	-----	----	--------	----	-----	------	----	-----------

```
## 8.021589 9.599323 9.955285 10.34744 11.81257 9.955656 0.5697085 1000 0
```

- Our bootstrapped estimate of the mean would be 9.96.
- We can create a 95% confidence interval for this point estimate by eliminating the smallest/largest 2.5% of the Tstar values (this is called the percentile bootstrap confidence interval). The smallest value that is left is the lower bound on the confidence interval and the largest value that is left is the upper bound on the confidence interval.

```
Tstar_trimmed <- sort(Tstar)[26:975]
min(Tstar_trimmed)
```

```
## [1] 8.793786
```

```
max(Tstar_trimmed)
```

```
## [1] 11.02616
```

We can also calculate the bootstrap bias.

$$\widehat{\text{Bias}}_{\text{boot}} = \bar{T}^* - \hat{\theta}$$

where $\hat{\theta}$ is the statistic computed from the original sample and $\bar{T}^*$ is the mean of the bootstrap statistics.

```
Bias <- mean(Tstar) - mean(xsample)
Bias
```

```
## [1] -0.006525734
```

In general, the bootstrapped mean is unbiased, but it is conditional on the observed sample. This means that the randomness being averaged over comes only from the resampling process, while the original data are treated as fixed. The bootstrap treats the observed data as fixed and only averages over the resampling process, not over new samples from the population.

This result is special to linear statistics like the mean. For many nonlinear statistics (e.g., variance, median, correlation, maximum), the bootstrap estimate is generally biased, often biased upward, and the bootstrap is frequently used to estimate and correct that bias.

Because we know that the population is normal, we can create a t-interval in order to see how well the bootstrap procedure worked.

```
myinterval <- t.test(xsample)
myinterval$conf.int
```

```
## [1] 8.788536 11.135827
```

```
## attr(,"conf.level")
```

```
## [1] 0.95
```

There are several bootstrap confidence intervals, including the percentile (the one I used), basic, studentized, and BCa intervals. These differ in how they correct for bias and skewness in the bootstrap distribution.

When the bootstrap fails

At minimum, note that the bootstrap can perform poorly when:

- Sample size is very small
- Statistic is not smooth (e.g., sample maximum, quantiles at boundaries)
- Data are dependent (time series, spatial data)
- Heavy tails or extreme skewness

In summary, the bootstrap can be viewed as a numerical analogue of the delta method, replacing analytic approximations with simulation.

Jackknife

The jackknife is a resampling method that actually predates the bootstrap. It is mainly used for estimating the variance and bias of an estimator. Maurice Quenouille actually invented the concept, but in 1958 John Tukey expanded on the idea and coined the name jackknife since it is “is a rough-and-ready tool that can improvise a solution for a variety of problems even though specific problems may be more efficiently solved with a purpose-designed tool.”(Tukey, John W. (1958). “Bias and confidence in not quite large samples (abstract)”. The Annals of Mathematical Statistics. 29 (2): 614.)

The jackknife estimate of the mean is obtained (for a sample of size n) by computing n means by leaving each sample out.

For example, the i^{th} mean is obtained by the following formula:

$$\bar{x}_{(i)} = \frac{\sum_{j=1}^{i-1} x_j + \sum_{j=i+1}^n x_j}{n-1} = \frac{n\bar{x} - x_i}{n-1}$$

The mean is calculated without the i^{th} data point.

$$\bar{x}_{\text{jack}} = \frac{1}{n} \sum_{i=1}^n \frac{n\bar{x} - x_i}{n-1} = \frac{1}{n(n-1)} \left(n^2\bar{x} - \sum_{i=1}^n x_i \right)$$

Notice that because $\sum_{i=1}^n x_i = n\bar{x}$,

$$\bar{x}_{\text{jack}} = \frac{1}{n(n-1)} (n^2\bar{x} - n\bar{x}) = \frac{n\bar{x}(n-1)}{n(n-1)} = \bar{x}$$

The jackknife estimate of the mean is exactly equal to the sample mean — not asymptotically, but identically.

Since we know that the sample mean is unbiased and it's equal to the jackknife estimate of the mean, we know that the jackknife estimate of the mean is unbiased. However, not all estimators are unbiased, and the jackknife can be used to estimate and correct bias in nonlinear estimators → We can use the jackknife as a technique for estimating bias in nonlinear estimators.

Let $\hat{\theta}$ be any jackknife estimator (e.g. mean, median).

$$\hat{\theta}_{\text{jack}} = \frac{\hat{\theta}_{(1)} + \hat{\theta}_{(2)} + \dots + \hat{\theta}_{(n)}}{n} = \frac{1}{n} \sum_{i=1}^n \hat{\theta}_{(i)}$$

where $\hat{\theta}_{(i)}$ is the estimate of theta with the i^{th} point removed.

Example: Given the data set 1,3,4,5,7,7,10,11. Find the jackknife estimate of the median.

$$\theta_{(1)} = \text{median}(3, 4, 5, 7, 7, 10, 11) = 7, \theta_{(2)} = \text{median}(1, 4, 5, 7, 7, 10, 11) = 7$$

$$\theta_{(3)} = \text{median}(1, 3, 5, 7, 7, 10, 11) = 7, \theta_{(4)} = \text{median}(1, 3, 4, 7, 7, 10, 11) = 7$$

$$\theta_{(5)} = \text{median}(1, 3, 4, 5, 7, 10, 11) = 5, \theta_{(6)} = \text{median}(1, 3, 4, 5, 7, 10, 11) = 5$$

$$\theta_{(7)} = \text{median}(1, 3, 4, 5, 7, 7, 11) = 5, \theta_{(8)} = \text{median}(1, 3, 4, 5, 7, 7, 10) = 5$$

$$\hat{\theta}_{\text{jack}} = \frac{7 + 7 + 7 + 7 + 5 + 5 + 5 + 5}{8} = 6$$

Bias of the Jackknife

The jackknife approximates how the estimator changes under small perturbations of the empirical distribution (a first-order approximation).

$$\widehat{\text{Bias}}(\hat{\theta})_{\text{jack}} = (n-1)(\hat{\theta}_{\text{jack}} - \hat{\theta})$$

The factor $(n-1)$ rescales the leave-one-out perturbation, which is of order $1/n$, to recover the full $O(n^{-1})$ bias of $\hat{\theta}$. Removing one observation changes the estimator by approximately $1/(n-1)$ of its total bias, so multiplying by $n-1$ reconstructs the full bias.

Bias Corrected Jackknife

$$\hat{\theta}_{JK} = \hat{\theta} - \widehat{\text{Bias}}(\hat{\theta})_{\text{jack}} = n\hat{\theta} - (n-1)\hat{\theta}_{\text{jack}}$$

Why the jackknife bias correction works (pseudovalue interpretation)

The bias-corrected jackknife estimator can be understood through the *pseudovalue* representation:

$$\theta_i^* = n\hat{\theta} - (n-1)\hat{\theta}_{(i)},$$

which rescales the effect of removing a single observation from the estimator. Under standard regularity conditions, these pseudovalues behave approximately like independent and identically distributed observations whose mean is close to the target parameter θ . The jackknife bias correction and variance estimator then arise naturally as the mean and dispersion of the pseudovalues, providing a first-order correction for estimators whose bias is $O(n^{-1})$.

Jackknife estimate of the variance

Since the pseudovalues are approximately iid., the variance of $\hat{\theta}$ (i.e. the variance of the estimator using the entire sample) may be estimated by the sample variance of the pseudovalues divided by n :

$$\widehat{\text{Var}}_{JK}(\hat{\theta}) = \frac{1}{n(n-1)} \sum_{i=1}^n (\theta_i^* - \bar{\theta}^*)^2, \quad \bar{\theta}^* = \frac{1}{n} \sum_{i=1}^n \theta_i^*.$$

This expression is algebraically equivalent to

$$\widehat{\text{Var}}_{JK}(\hat{\theta}) = \frac{n-1}{n} \sum_{i=1}^n (\hat{\theta}_{(i)} - \hat{\theta}_{JK})^2.$$

This differs from a standard sample variance because the leave-one-out estimates $\hat{\theta}_{(i)}$ are highly dependent and have variability of order n^{-1} rather than $n^{-1/2}$. The factor $\frac{n-1}{n}$ rescales this artificially small dispersion so that, under asymptotic linearity of $\hat{\theta}$, the resulting estimator consistently estimates $\widehat{\text{Var}}_{JK}(\hat{\theta})$.

Relationship between jackknife variance and bias:

Let $\hat{\theta}$ be an estimator with leave-one-out versions $\hat{\theta}_{(i)}$. The jackknife bias estimator is

$$\widehat{\text{Bias}}_{JK}(\hat{\theta}) = (n-1)(\hat{\theta}_{\text{jack}} - \hat{\theta}), \quad \hat{\theta}_{\text{jack}} = \frac{1}{n} \sum_{i=1}^n \hat{\theta}_{(i)}$$

and the jackknife variance estimator is

$$\widehat{\text{Var}}_{JK}(\hat{\theta}) = \frac{n-1}{n} \sum_{i=1}^n (\hat{\theta}_{(i)} - \hat{\theta}_{\text{jack}})^2$$

Both the jackknife bias and variance estimators are derived from the same set of pseudovales $\{\theta_i^*\}_{i=1}^n$. The bias estimator depends on the *mean* of the pseudovales relative to $\hat{\theta}$, while the variance estimator depends on their *dispersion*. Thus, bias and variance are different summaries of the same first-order sensitivity of $\hat{\theta}$ to individual observations.

The jackknife estimate of the variance in the previous example would be:

$$\begin{aligned} \widehat{\text{Var}}_{JK}(\hat{\theta}) &= \frac{8-1}{8} [(7-6)^2 + (7-6)^2 + (7-6)^2 + (7-6)^2 + (5-6)^2 + (5-6)^2 \\ &\quad + (5-6)^2 + (5-6)^2] = \frac{7}{8}(8) = 7 \end{aligned}$$

$$\text{Jack SE} = \sqrt{7} = 2.645751$$

The jackknife bias of the previous example would be:

$$\widehat{\text{Bias}}_{JK}(\hat{\theta}) = (8-1)(6-6) = 0$$

When the jackknife fails

- Median

$$x = (0, 0, 0, 0, 1, 1, 1)$$

$\hat{\theta}$ is the median of all 7 values, which is 0.

If you delete one of the zeros (i=1,2,3,4), you get (0,0,0,1,1,1) and the median is (0+1)/2=0.5.

If you delete one of the ones, you get (0,0,0,0,1,1) and the median is 0.

$$\hat{\theta}_{\text{jack}} = \frac{0.5+0.5+0.5+0.5+0+0+0}{7} = 2/7$$

The bias correct version is,

$$\begin{aligned} \hat{\theta}_{JK} &= n\hat{\theta} - (n-1)\hat{\theta}_{\text{jack}} \\ &= 7(0) - (7-1)(2/7) = -12/7 = -1.714 \end{aligned}$$

This is outside of the range of the data points $\rightarrow$ non-sensical.

Key Insight: The jackknife fails for the median because the median is not a smooth functional of the data. Small changes in the sample can cause discrete jumps.

- Quantiles

The median is one of the quantiles, the same thing can happen with the other quantiles.

- Extreme order statistics

Consider a random sample of size 10 from $\text{Unif}(0,1)$. We want the jackknife estimate of the largest order statistic. Assume the largest 2 values are:

$$x_{(9)} = 0.8, x_{(10)} = 0.99$$

$$\hat{\theta} = 0.99$$

$$\hat{\theta}_{\text{jack}} = \frac{9(0.99) + 0.8}{10} = 0.971$$

$$\hat{\theta} = 0.99$$

The bias correct version is,

$$\hat{\theta}_{JK} = n\hat{\theta} - (n-1)\hat{\theta}_{\text{jack}}$$

$$= 10(0.99) - (10-1)0.971 = 1.161$$

This value is greater than 1 which is impossible if you sampled from $\text{Unif}(0,1)$.

- Estimators with discontinuities

→ The jackknife assumes a smooth functional of the empirical distribution.

Summary

Why/when would you use the bootstrap?

1. Bootstrapping makes fewer parametric assumptions, but assumes the empirical distribution is a good approximation to the population distribution. All you do is resample your original data and work with that distribution whatever that distribution turns out to be.
2. If your sample size is large enough, generally ≥ 30 and you are looking at the distribution of the mean, sometimes the central limit theorem can help us imply normality.
3. We can use bootstrapping to estimate a large variety of statistics. The mode, mean, median standard deviation, proportions, odds ratios. You can also use bootstrapping for variance statistical methods such as t-tests, Analysis of Variance, and estimating regression coefficients.
4. The bootstrap can fail or be unstable when the population variance is infinite or when the statistic is highly sensitive to extreme observations.

Why/when would you use the Jackknife?

1. Bootstrap is 10 times more computationally intensive than the jackknife. If computing resources are a problem, use jackknife.
2. Jackknife often produces slightly larger standard errors than bootstrap estimates.
3. The jackknife produces the same answer every time, whereas the bootstrap does not.
4. The jackknife performs best for smooth statistics (e.g., means, regression coefficients). It performs poorly for non-smooth statistics, such as extreme order statistics or quantiles at boundaries.
5. The jackknife can be useful for small samples when the estimator is smooth, but it relies on asymptotic arguments and may perform poorly for nonsmooth estimators.

Method	Strength	Weakness
Bootstrap	General, flexible	Computationally intensive
Jackknife	Fast, simple	Fails for non-smooth stats